

A unified account of multiple homophony of *m*- in Seediq¹

21 June 2024 | 16-ICAL

Yvette Yi-Chi Wu | yichiwu@g.harvard.edu | Harvard University

- the **prefix *m*-** appears to serve multiple functions in Seediq (1) (adapted from Chen 2023)
 - at the same time, the **actor voice (AV) marker** alternates between prefix *m*- and infix $\langle m \rangle$ (2)
- (1)
- | | | | |
|----|---------------|--|--|
| a. | actor voice: | <i>m-ekan</i> ‘eat’, <i>m-osa</i> ‘go’ | (vs. C-initial <i>t\langle m \rangle alang</i> ‘run’) |
| b. | detransitive: | <i>m-gerung</i> ‘break (intr.)’ | (vs. transitive <i>g\langle m \rangle erung</i> ‘break’) |
| c. | stative: | <i>m-narux</i> ‘be sick’ | |
| d. | reciprocal: | <i>m-g-gaalu</i> ‘care for each other’ | (vs. transitive <i>g\langle m \rangle aalu</i> ‘care’) |
- (2)
- | | | |
|----|---------------|--|
| a. | unaccusative: | <i>m-rkerak</i> ‘itch’, <i>m-huqil</i> ‘die’, <i>m-seyaq</i> ‘cough’, <i>m-eniq</i> ‘exist’ |
| b. | unergative: | <i>t\langle m \rangle alang</i> ‘run’, <i>r\langle m \rangle engo</i> ‘talk’, <i>l\langle m \rangle ingis</i> ‘cry’, <i>m-eyah</i> ‘come’ |
| c. | transitive: | <i>c\langle m \rangle hungi</i> ‘forget’, <i>s\langle m \rangle ipaq</i> ‘kill’, <i>d\langle m \rangle urun</i> ‘ask’, <i>m-angal</i> ‘take’ |
- this talk proposes a unified account of the instances above of /m/ as a single allomorph of the AV marker, with the underlying form as an **infix** inserted before the first vowel of the stem
 - i.e. the vocabulary insertion rule is: AV $\Leftrightarrow \langle m \rangle$, with infix location of $_\text{V}$
 - additional morphosyntactic material intervenes between the AV marker and the verb root
 - builds on Ross 1995; Kaufman 2012; Chen 2020 for **bimorphemic approaches** to AV variants
 - various **argument structure related affixes** (e.g. detransitive *V-*, stative *k-*, or reciprocal *p-*) are underlyingly present but obscured due to regular phonological processes (Holmer 1996)
 - these processes give rise to the appearance of a prefix *m*- despite underlying morpheme breakdown

main claims:

- the alternation between actor voice (AV) prefix *m*- and infix $\langle m \rangle$ is phonologically predictable
- the prefix *m*- in anticausative, statives, and reciprocals (Chen 2023) contains a regular AV marker
- the **order of affixes** in Seediq is: Asp_{PFV} > “voice” > Voice_[±D] > Asp_{RED} > Appl_{CAUS} > *v*
 - NOTE: this is based on the ordering of left-aligned voice affixes such as AV and IV

1 basic Seediq facts

1.1 Austronesian voice

- **voice and aspect morphology** reconstructs to Proto-Austronesian (PAN) (Blust 2009):
 - actor voice (AV) $\langle m \rangle$, patient voice (PV) *-un*, locative voice (LV) *-an*, instrumental voice (IV) *s-*

¹my greatest thanks to language teachers and friends pai Habo, pai Lubi, Takun ptgesa, Pihu ptgesa, Mahung, Temu, Rabe, Pawan, Temi, Uma ptgesa, Tado ptgesa, and Obing ptgesa, as well as Jonathan Bobaljik, Laura Kalin, Lisa Travis, Heather Newell, and audiences at the Harvard WOLF Lab for their helpful discussions; unless otherwise noted, Seediq data throughout this paper comes from primary fieldwork conducted in Alang Gluban in Taiwan from 2022 to 2024, with the Tgdaya dialect of Seediq

- perfective aspect or past tense *⟨in⟩; imperfective or prospective aspect *RED_{CV}-
- Seediq preserves this morphology, though some functions have shifted (Holmer 1996; Tsukida 2005)
 - AV and PV alternations are still productive for promotion of agent vs. patient (3)
 - LV is often used for patient promotion, and IV is not productive with TAM morphology

- (3) a. wada **m**-ita arung [ka Habo]
 PST AV-see pangolin PIV Habo
 ‘Habo saw a pangolin.’ actor voice (AV)
- b. wada qta-**un** na Habo [ka arung]
 PST see-PV GEN Habo PIV pangolin
 ‘Habo saw the pangolin.’ patient voice (PV)

- the full **paradigm** of Seediq verbal morphology is shown below (4) (adapted from Holmer 1996)²
 - AV and IV are left-aligned affixes while PV and LV are right-aligned affixes
 - voice is mostly compositional with tense/aspect but has portmanteau forms with mood

(4)

	AV	PV	LV	IV
PRES	m- or ⟨m⟩	-un	-an	s-
PFV	m-n- or ⟨m-n⟩	⟨n⟩	⟨n⟩ -an	-
PSV	m-RED- or RED⟨m⟩-	RED- -un	RED- -an	-
FUT	m-p-	RED- -un	RED- -an	-
IMP	∅	-i	-an-i	-
SUBJ	m- -a or ?	-o	-e	-

1.2 phonological processes

- Seediq has undergone **reduction** of pre-stress vowels (5) (Yang 1976; Holmer 1996)
 - all vowels in pre-penultimate syllables are phonologically reduced to an [u] or [ə]
 - exception #1: pre-stress vowels will match the following vowel if separated by an /ʔ/ (5d,e)
 - exception #2: pre-stress vowels are elided if preceded by a non-initial /m/ or /n/ (5f)

- in the orthography, all pre-stress vowels are omitted unless followed by a glottal stop

- (5) a. *tg-daya* ‘up the hill’ → [tugu-daya] d. *seediq* ‘person’ → [se-ʔediq]
 b. *s⟨m⟩ipaq* ‘⟨AV⟩kill’ → [su-mipaq] e. *kn-teetu* ‘often’ → [kunte-ʔetu]
 c. *qbheni* ‘bird’ → [qubu-heni] f. *s⟨m⟩kuxul* ‘⟨AV⟩like’ → [sum-kuxul]

- **consonant deletion** occurs when labial and back consonants are followed by /m/ (6) (Holmer 1996)

- (6) a. *p⟨m⟩atis* ‘⟨AV⟩write’ → [matis] c. *q⟨m⟩ita* ‘⟨AV⟩see’ → [mita]
 b. *b⟨m⟩ege* ‘⟨AV⟩buy’ → [mege] d. *k⟨m⟩hemuc* ‘⟨AV⟩at.will’ → [muhemuc]

²the glossing here follows the Leipzig glossing conventions, with the addition of the following: PIV = pivot, AV = actor voice, PV = patient voice, LV = location voice, IV = instrumental voice, ANTICAUS = anticausative, PSV: prospective aspect

2 prefix *m-*

2.1 actor voice (AV)

- at first glance, the **alternation** of AV morphology as the prefix *m-* vs. infix $\langle m \rangle$ in Seediq is sensitive to both phonological and syntactic information, with the generalization summarized in (7):
 - agentive verbs (e.g. unergatives and transitives) take as a baseline the infix $\langle m \rangle$ (8b,c)
 - non-agentive verbs (e.g. unaccusative and statives) take as a baseline the prefix *m-* (8a)
 - however, infix $\langle m \rangle$ will surface as a prefix *m-* with all vowel-initial (or /ʔ/-initial) roots³
- this generalization has been described in terms of **verb classes** in PAn (Tsuchida 1976; Ross 2015)

- (7) a. C-initial *and* agentive roots → $\langle m \rangle$
 b. elsewhere → *m-*
 c. idiosyncratic (often roots with three or more syllables) → ∅
- (8) a. unaccusative: *m-rkerak* ‘itch’, *m-sngari* ‘remain’, *m-huqil* ‘die’, *m-seyaq* ‘cough’, *m-takur* ‘fall’, *m-eniq* ‘exist’, *m-iicuw* ‘be scared’, *dehuk* ‘arrive’, *bsukan* ‘be drunk’, *tkriyak* ‘appear’
 b. unergative: *s⟨m⟩rebu* ‘poop’, *t⟨m⟩alang* ‘run’, *r⟨m⟩engo* ‘talk’, *l⟨m⟩ingis* ‘cry’, *k⟨m⟩eeke* ‘dance’, *m-uuyas* ‘sing’, *m-eyah* ‘come’, *m-alix* ‘give up’, *skiya* ‘fly’, *qduriq* ‘flee’
 c. transitive: *g⟨m⟩beling* ‘dig’, *c⟨m⟩hungi* ‘forget’, *s⟨m⟩ipaq* ‘kill’, *k⟨m⟩eruc* ‘cut’, *c⟨m⟩ebu* ‘shoot’, *d⟨m⟩urun* ‘ask’, *m-ekan* ‘eat’, *m-angal* ‘take’, *m-etur* ‘kick’, *hpure* ‘cook’, *snhii* ‘believe’

- this distinction is **not tied** to lexical items, as roots can occur with both allomorphs (9) (Chen 2020)
 - transitive forms with infix $\langle m \rangle$ vs. their reflexive/detransitive counterparts with prefix *m-*

- (9) a. *t⟨m⟩utuy=ku* *heya*
 $\langle AV \rangle$ wake=1SG.PIV 3.SG
 ‘I am waking her up.’ (Holmer 1999)
 b. *m-tutuy laqi=su*
 AV-wake child=2SG.GEN
 ‘Your child is getting up.’ (Holmer 1999)
- c. *wada d⟨m⟩engu qhuni ka Dakis*
 PST $\langle AV \rangle$ roast wood PIV Dakis
 ‘Dakis heated the wood.’ (Chen 2020)
 d. *wada m-dengu ka qhuni*
 PST AV-roast PIV wood
 ‘The wood has been heated.’ (Chen 2020)

2.2 other functions

- the prefix *m-* can be attributed to several historically independent functions in PAn (Chen 2023)
 - actor voice **m-*, stative **ma-*, detransitive **mu-*, and reciprocal **maCa-* (10)
 - due to the reduction of pre-stress vowels (§1.2), this difference in vowels is obscured (11)

- (10) a. actor voice: PAn **m-*, * $\langle um \rangle$ c. detransitive: PAn **mu-*
 b. stative: PAn **ma-* d. reciprocal: PAn **maCa-*
- (11) a. *m-ekan* ‘eat’ → [m-ekan] c. *m(u)-gerung* ‘break (intr.)’ → [mu-gerung]
 b. *m(a)-narux* ‘be sick’ → [mu-narux] d. *m(a)-g(a)-gaalu* ‘care for e.o.’ → [mu-gu-gaʔalu]

³see §5.3 for evidence that we need both vowel-initial and /ʔ/-initial roots, and one cannot be reduced to the other

- (12) a. actor voice: *m-ekan* ‘eat’, *m-eyah* ‘come’, *m-osa* ‘go’, *m-angal* ‘take’, *m-aha* ‘go’
 b. stative: *m-narux* ‘be sick’, *m-sekuy* ‘be cold’, *m-tanah* ‘be red’, *mi-icuw* ‘be scared’
 c. detransitive: *m-gerung* ‘break’, *m-slamaq* ‘burn’, *m-temaq* ‘crumble’, *m-takur* ‘fall’
 d. reciprocal: *m-g-gaalu* ‘care for e.o.’, *m-b-bege* ‘give e.o.’, *m-d-dayo* ‘help e.o.’

- however, note that while what is orthographically the prefix *m-* does genuinely correspond to the consonant [m-] for the AV marker, it instead corresponds to the CV sequence [mu-] for the others
- this hints at the additional morphosyntactic material that intervenes between AV *m-* and the root

3 proposal

- I expand on bimorphemic approaches to AV variants in Austronesian languages (Ross 1995; Kaufman 2012; Teng 2014; Chen 2020, 2022) and argue that the affixes in (10) are all decomposable
- I bring in new evidence based on interactions of these affixes with negation and TAM morphology

proposal #1: the instances of /m/ are all derived from a **single allomorph** of the AV marker, the infix ⟨m⟩

- the infix ⟨m⟩ is left-aligned and inserted before the first vowel of the stem (not the root!) (13)
 - see Yu (2003); Kalin (2022) for the placement of infixes w.r.t. phonological or prosodic properties
- for regular verbs, this surfaces as 1) an **infix** with C-initial roots and 2) a **prefix** with V-initial roots

(13) vocabulary item for actor voice: AV ⇔ ⟨m⟩ with infix location of _V

- (14) a. C-initial stem: *talang* ‘run’ → *t⟨m⟩alang* ‘⟨AV⟩run’ (/m/ goes before first vowel /a/)
 b. V-initial stem: *osa* ‘go’ → ⟨m⟩*osa* ‘⟨AV⟩go’ (/m/ goes before first vowel /o/)

proposal #2: the apparent cases of the prefix *m-* with C-initial roots is due to additional morphosyntactic material that intervenes between the AV marker and the verb root, that is **phonologically reduced**

- previous work: 1) **mu-* decomposes into AV **m-* and detransitive **u-* (Chen 2020)
 2) **ma-* decomposes into AV **⟨um⟩* and stative **ka-* (Ross 1995; Kaufman 2012)

- (15) vocabulary items: combined with AV infix ⟨m⟩:
- | | | | |
|---------|----------------------------|-------------------|---------|
| a. DETR | ⇔ V- | ⟨m⟩V- ‘⟨AV⟩DETR’ | → [mu-] |
| b. REC | ⇔ pV- (with reduplication) | p⟨m⟩V- ‘⟨AV⟩REC’ | → [mu-] |
| c. STAT | ⇔ kV- | k⟨m⟩V- ‘⟨AV⟩STAT’ | → [mu-] |

- the **consonants** in these argument structure related affixes are systematically **deleted** (§1.2)
- because roots in Seediq are at a minimum bisyllabic, the vowels in these morphemes will always be located in a pre-penultimate syllable and be reduced as [u], altogether resulting in the form [mu-]

proposal #3: the **order of affixes** in Seediq is: Asp_{PFV} > “voice” > Voice_[±D] > Asp_{RED} > Appl_{CAUS} > v

- assuming that DETR and REC are Voice heads (Labelle 2008; Schafer 2008) and STAT is a v head
- see Travis 2010 for reasons that 1) **outer aspect** and 2) **inner aspect** are distinct in Austronesian, where the latter is reduplication that sits below Voice and conveys an incomplete/ongoing action

4 infixation of AV

4.1 argument structure

- overall, all that is needed to account for the data are the following **vocabulary items** (16):
 - the **Voice_[-D]** head that spells out V- occurs more broadly in detransitive and unaccusative contexts (Chen 2022), which are unified by the lack of an external argument (Kastner 2017)
 - not shown below: the regular agentive Voice_[+D] head and active *v* head are null (∅)

(16)	vocabulary items:	head which hosts it:
a.	AV ⇔ ⟨ <i>m</i> ⟩ with infix location of __V	“voice”
b.	DETR ⇔ V-	Voice _[-D]
c.	REC ⇔ <i>p</i> V- with reduplication	Voice _[+D]
d.	STAT ⇔ <i>k</i> V-	<i>v</i>

- the transformation of these underlying forms into their surface output is done via **regular phonological processes** in the language (§1.2), namely vowel reduction and consonant deletion
 - the combination of the AV and DETR/REC/STAT markers results in the form [mu-] (17)
 - the presence of Voice_[-D] in unaccusatives results in the prefix *m*- vs. infix ⟨*m*⟩ alternation (18)

(17)		morpheme gloss:	underlying form:	phonological output:
a.	detransitive:	‘⟨AV⟩DETR-break’	⟨ <i>m</i> ⟩ <i>V-gerung</i>	[mu-gerung]
b.	reciprocal:	‘⟨AV⟩REC-care.for’	<i>p</i> ⟨ <i>m</i> ⟩ <i>V-ga-gaalu</i>	[mugu-gaalu]
c.	stative:	‘⟨AV⟩STAT-be.sick’	<i>k</i> ⟨ <i>m</i> ⟩ <i>V-narux</i>	[mu-narux]

(18)		morpheme gloss:	underlying form:	phonological output:
a.	unaccusative:	‘⟨AV⟩UNACC-die’	⟨ <i>m</i> ⟩ <i>V-huqil</i>	[mu-huqil]
b.	unergative:	‘⟨AV⟩run’	<i>t</i> ⟨ <i>m</i> ⟩ <i>alang</i>	[tu-malang]
c.	transitive:	‘⟨AV⟩forget’	<i>c</i> ⟨ <i>m</i> ⟩ <i>hungi</i> ⁴	[cumu-hungi]

- additional evidence comes from the difference in behavior of vowel-initial verb roots:
 - complex (DETR/REC/STAT) verbs occur with an **extra vowel** between *m*- and the root (19)
 - in contrast, simple agentive verbs have the prefix *m*- attach directly to the root (20)
- (19) a. *uwic* ‘be.tired’ → *kV-uwic* ‘STAT-be.tired’ → *k*⟨*m*⟩*V-uwic* ‘⟨AV⟩STAT-be.tired’ → *mu-uwic*
 b. *icuw* ‘be.scared’ → *kV-icuw* ‘STAT-be.scared’ → *k*⟨*m*⟩*V-icuw* ‘⟨AV⟩STAT-be.scared’ → *mi-icuw*
- (20) a. *osa* ‘go’ → *osa* ‘go.Voice_[+D]’ → ⟨*m*⟩*osa* ‘⟨AV⟩go.Voice_[+D]’ → *m-osa*
 b. *angal* ‘take’ → *angal* ‘forget.Voice_[+D]’ → ⟨*m*⟩*angal* ‘⟨AV⟩take.Voice_[+D]’ → *m-angal*

⁴I assume that roots underlyingly have vowels for every syllable despite later vowel reduction, hence with tri-syllabic roots like *chungi* ‘forget’ (underlyingly *cVhungi*), AV infixes between the first and second consonants, resulting in *cmhungi*

4.2 negation

- if the consonants in these morphemes are systematically deleted, how do we know they are there?
 - instances where AV ⟨*m*⟩ does not occur and/or does not trigger deletion
 - constructions: 1) negative forms, 2) PV and LV forms, 3) additional intervening material
- negative (*aka* connegative) AV forms **lack overt voice morphology** in Seediq (21) (Holmer 1996)
 - the affirmative ~ negative alternation reveals the underlying consonants of /p/ and /k/ (22)
 - the expected alternations: 1) detransitive *m*- ~ ∅; 2) stative *m*- ~ *k*-; 3) reciprocal *m*- ~ *p*-

- (21)
- | | affirmative form: | | negative form: |
|-----------------|--|---|---------------------------------------|
| a. unaccusative | <i>m</i> - <i>huqil</i> ‘die’ | ~ | <i>ini</i> ∅- <i>huqil</i> ‘not die’ |
| b. unergative: | <i>t</i> (<i>m</i>)- <i>alang</i> ‘run’ | ~ | <i>ini</i> ∅- <i>talang</i> ‘not run’ |
| c. transitive: | <i>m</i> - <i>ekan</i> ‘eat’ | ~ | <i>ini</i> ∅- <i>ekan</i> ‘not eat’ |
- (22)
- | | affirmative form: | | negative form: |
|------------------|--|---|---|
| a. detransitive: | <i>m</i> - <i>gerung</i> ‘break’ | ~ | <i>ini</i> ∅- <i>gerung</i> ‘not break’ |
| b. reciprocal: | <i>m</i> - <i>g-gaalu</i> ‘care for e.o.’ | ~ | <i>ini</i> <i>p</i> - <i>g-gaalu</i> ‘not care for e.o.’ |
| c. stative: | <i>m</i> - <i>narux</i> ‘be sick’ | ~ | <i>ini</i> <i>k</i> - <i>narux</i> ‘not be sick’ |
- (23)
- detransitive: ***m***-*slamaq* ~ *ini* ∅-*slamaq* ‘burn’, ***m***-*temaq* ~ *ini* ∅-*temaq* ‘crumble’
 - reciprocal: ***m***-*c-cebu* ~ *ini* ***p***-*c-cebu* ‘shoot e.o.’, ***m***-*d-dayo* ~ *ini* ***p***-*d-dayo* ‘help e.o.’, ***m***-*b-bege* ~ *ini* ***p***-*b-bege* ‘give e.o.’, ***m***-*k-kela* ~ *ini* ***p***-*k-kela* ‘know e.o.’, and more
 - stative: ***mi***-*icuw* ~ *ini* ***ki***-*icuw* ‘be scared’, ***m***-*sekuy* ~ *ini* ***k***-*sekuy* ‘be cold’, ***m***-*taqi* ~ *ini* ***q***-*taqi* ‘sleep’, ***m***-*seyaq* ~ *ini* ***q***-*seyaq* ‘cough’, ***m***-*dengu* ~ *ini* ***k***-*dengu* ‘be dry’, and more

- it is not exactly predictable which roots participate in which **argument structure alternations**
 - e.g. active vs. stative (24a), transitive vs. detransitive (24b), causative vs. inchoative (24c)
 - transitive roots are not productively compatible with detransitive morphology as in some other languages, and is reserved for prototypical anticausatives (Schafer 2008) (25)

- (24)
- | | active/transitive/causative form: | | stative/detransitive form: |
|----|--|---|--|
| a. | <i>t</i> (<i>m</i>)- <i>akur</i> ~ <i>ini</i> ∅- <i>takur</i> ‘trip’ | ↔ | <i>m</i> - <i>takur</i> ~ <i>ini</i> <i>k</i> - <i>takur</i> ‘fall’ |
| | <i>d</i> (<i>m</i>)- <i>engu</i> ~ <i>ini</i> ∅- <i>dengu</i> ‘roast’ | ↔ | <i>m</i> - <i>dengu</i> ~ <i>ini</i> <i>k</i> - <i>dengu</i> ‘be dry’ |
| b. | <i>s</i> (<i>m</i>)- <i>lamaq</i> ~ <i>ini</i> ∅- <i>slamaq</i> ‘burn (tr.)’ | ↔ | <i>m</i> - <i>slamaq</i> ~ <i>ini</i> ∅- <i>slamaq</i> ‘burn (intr.)’ |
| | <i>g</i> (<i>m</i>)- <i>erung</i> ~ <i>ini</i> ∅- <i>gerung</i> ‘break (tr.)’ | ↔ | <i>m</i> - <i>gerung</i> ~ <i>ini</i> ∅- <i>gerung</i> ‘break (intr.)’ |
| c. | <i>p</i> - <i>ki-icuw</i> ~ <i>ini</i> <i>p</i> - <i>ki-icuw</i> ‘scare’ | ↔ | <i>mi</i> - <i>icuw</i> ~ <i>ini</i> <i>ki</i> - <i>icuw</i> ‘be scared’ |
| | <i>p</i> - <i>k-narux</i> ~ <i>ini</i> <i>p</i> - <i>k-narux</i> ‘make sick’ | ↔ | <i>m</i> - <i>narux</i> ~ <i>ini</i> <i>k</i> - <i>narux</i> ‘be sick’ |
- (25)
- | | | | |
|----|--|----|--|
| a. | <i>m</i> - <i>ekan</i> ‘eat’ ↔ * <i>me</i> - <i>ekan</i> ‘be eaten’ | c. | <i>q</i> (<i>m</i>)- <i>erac</i> ‘catch’ ↔ * <i>m</i> - <i>qerac</i> ‘be caught’ |
| b. | <i>b</i> (<i>m</i>)- <i>ari</i> ‘buy’ ↔ * <i>m</i> - <i>bari</i> ‘be bought’ | d. | <i>k</i> (<i>m</i>)- <i>eruc</i> ‘forget’ ↔ * <i>m</i> - <i>keruc</i> ‘be cut’ |

- under preliminary observation, Seediq is undergoing a **merger** of the detransitive/reciprocal/stative morphemes as *V*-, where /p/ and /k/ fail to appear in negative constructions in younger generations

5 affix ordering

- the **order of affixes** in Seediq is: $\text{Asp}_{\text{PFV}} > \text{“voice”} > \text{Voice}_{[\pm\text{D}]} > \text{Asp}_{\text{RED}} > \text{Appl}_{\text{CAUS}} > v$
 - NOTE: this is based off of the ordering of left-aligned voice affixes such as AV and IV
 - potentially at odds with theories where voice is structurally high (i.e. A' domain)
- additional **vocabulary items** that are relevant to this ordering are the following (26):
 - assuming that PFV/PSV are outer/inner Asp heads (Travis 2010), and CAUS is an Appl head

(26)	vocabulary items:			head which hosts it:
a.	PFV	$\Leftrightarrow$	$\langle n \rangle$ with infix location of __V	Asp_{PFV}
b.	PSV	$\Leftrightarrow$	$\text{RED}_{\text{CV-}}$	Asp_{RED}
c.	CAUS	$\Leftrightarrow$	$pV-$	$\text{Appl}_{\text{CAUS}}$

5.1 “voice”

- I use the term “voice” descriptively to refer to AV/PV/LV/IV morphology, not to be confused with the external argument introducing head Voice (Kratzer 1996); what “voice” is will be discussed later
- “**voice**” is ordered **above** argument structure affixes (as seen in §4.1), which includes Voice and v
 - see Chen 2022 as well for “voice” being hosted above $\text{Voice}_{[-\text{D}]}$ in Puyuma
 - otherwise, AV would infix into C-initial roots regardless of argument structure affixes (27a,b), and AV would directly prefix onto V-initial roots without an extra intervening vowel (27c)

(27)		“voice” > Voice, v	*Voice, v > “voice”
a.	<i>gerung</i> ‘break’:	$\langle m \rangle V\text{-gerung}$ [mu-gerung]	* $V\text{-g}\langle m \rangle \text{erung}$ [gu-merung]
b.	<i>cebu</i> ‘shoot’:	$p\langle m \rangle V\text{-ca-cebu}$ [mugu-cebu]	* $pV\text{-c}\langle m \rangle a\text{-cebu}$ [pucumu-cebu]
c.	<i>uwic</i> ‘be tired’:	$k\langle m \rangle V\text{-uwic}$ [mu-?uwic]	* $kV\text{-}\langle m \rangle \text{uwic}$ [ku-muwic]

- there is **root suppletion** conditioned by “voice” morphology, as seen in AV vs. PV forms (28)
 - given that suppletion is locally conditioned (Bobaljik 2012), this suggests that “voice” morphology must be local to and lie within the same maximal projection (i.e. **phase**) of the verb root
 - some PV forms have historical causative morphology that now appears to be grammaticalized

(28)		AV root	AV form	PV root	PV form
a.	‘give’:	<i>bege</i>	$b\langle m \rangle \text{ege}$	<i>biq</i>	<i>biq-un</i>
b.	‘eat’:	<i>ekan</i>	$m\text{-ekan}$	<i>p-uq</i>	<i>p-uq-un</i> (with CAUS)
c.	‘follow’:	<i>snegul</i>	$s\langle m \rangle \text{negul}$	<i>p-snol</i>	<i>p-snol-un</i> (with CAUS)
d.	‘like’:	<i>skuxul</i>	$s\langle m \rangle \text{kuxul}$	<i>kux</i>	<i>kux-un</i>

- suppletion seems to correspond to how affixation affects stress and vowel reduction in roots
 - two sets: 1) left-aligned affixes AV $\langle m \rangle$ and IV $s-$ and 2) right-aligned affixes PV $-un$ and LV $-an$
 - however, suppletion cannot be reduced to regular phonology, because other roots do not undergo the same process, e.g. $t\langle m \rangle \text{ekan} \sim t\text{kan-un}$ ‘pound’, $p\text{-bhege} \sim p\text{-bheg-un}$ ‘make white’

- **perfective aspect** (or past tense) PAn **<in>* is well attested in Austronesian languages (Blust 2009)
 - many languages have lost the infix **<in>* and replaced it with a prefix or enclitic
 - three Formosan languages preserve the perfective infix **<in>*: Bunun, Paiwan, and Seediq
- **Asp_{PFV}** is ordered **above** “voice” across Formosan languages
 - enclitics attach outside of voice (and in some languages outside of pronominal clitics) (29)
 - 1) PFV *<in>* infixes into AV and IV prefixes, or 2) PFV *na-* prefixes after the AV infix (30)

- (29) a. tu=idap-**ay**=**lra** [na sawka] ni Risem
3SG.GEN=sweep-LV=PFV PIV kitchen GEN Risem
'Risem swept the house.'
[Puyuma; Zheng 2018]
- b. kuna-**an**=na=**ti** [ya sunis]
first-PV=3SG.GEN=PFV PIV child
'He surpassed the child.'
[Kavalan; Chang 2006]

- (30) a. Bunun (NTU 2016): *m*⟨*i*⟩-*ludah* ‘AV⟨PFV⟩-hit’ (vs. *l*⟨*in*⟩*udah* ‘⟨PV.PFV⟩hit’)
s⟨*in*⟩-*pit’ia* ‘IV⟨PFV⟩-cook’ (vs. *p*⟨*in*⟩*it’ia* ‘⟨PV.PFV⟩cook’)
- b. Paiwan (Chang 2018): *na-t*⟨*em*⟩*ekel* ‘PFV-⟨AV⟩drink’ *na-k*⟨*em*⟩*an* ‘PFV-⟨AV⟩eat’
s⟨*in*⟩*i-kelem* ‘IV⟨PFV⟩-hit’ *s*⟨*in*⟩*i-patjavat* ‘IV⟨PFV⟩-leave’
- c. Seediq: ⟨*m-n*⟩*ekan* ‘⟨PFV⟩⟨AV⟩eat’ ⟨*m-n*⟩*imah* ‘⟨PFV⟩⟨AV⟩drink’
r⟨*m-n*⟩*engo* ‘⟨PFV⟩⟨AV⟩talk’ *l*⟨*m-n*⟩*ingis* ‘⟨PFV⟩⟨AV⟩cry’

- interestingly, Seediq is the only language where AV and PFV occur together as **infixes** (30c)
 - the ordering of AV and PFV is actually ambiguous depending on placement of the infix (31)

- | | | | |
|------|-------------------------|--|--|
| (31) | | Asp _{PFV} > “voice” / __V | *“voice” > Asp _{PFV} / __V |
| | | “voice” > Asp _{PFV} / C__ | *Asp _{PFV} > “voice” / C__ |
| a. | <i>rengo</i> ‘talk’: | <i>r</i> (m-n) <i>engo</i> [rum-nengo] | * <i>r</i> (n-m) <i>engo</i> [run-mengo] |
| b. | <i>osa</i> ‘go’: | m (n)- <i>osa</i> [mu-nosa] | * n (m)- <i>osa</i> [nu-mosa] |
| c. | <i>chungi</i> ‘forget’: | <i>c</i> (m-n) <i>hungi</i> [cumun-hungi] | * <i>c</i> (n-m) <i>hungi</i> [cunum-hungi] |

- I adopt the placement of $_\text{V}$, for consistency with the AV infix and other Formosan languages
 - both the infixes AV $\langle m \rangle$ and PFV $\langle n \rangle$ target the same location, before the first vowel of the stem, which creates the appearance that the PFV infix lies closer to the root than the AV infix (32)

- (32) a. *rengo* ‘talk’ → *r****m****engo* ‘⟨AV⟩talk’ → *r****m-n****engo* ‘⟨PFV⟩⟨AV⟩talk’ → *rmnengo*
 b. *osa* ‘go’ → ***m****osa* ‘⟨AV⟩go’ → ***m-n****osa* ‘⟨PFV⟩⟨AV⟩go’ → *mnosa*

- an **inner Aspect** projection has been proposed for Austronesian languages (Travis 2010)
 - this projection sits below Voice, and is realized through **CV reduplication**
 - reflects internal structure of an event, namely some kind of incomplete/ongoing action

- in Seediq, this projection is used for both **prospective aspect** (33a) and **reciprocals** (§4.1)
 - **Asp_{RED}** is ordered **below** “voice” across Formosan languages (Chen 2023)
 - AV ⟨*m*⟩ prefixes or infixes into the reduplicated material instead of the verb root (33)
- (33) a. *ciga m-e-ekan*=[ku] *ido m-eyah* Pawan
 yesterday AV-RED-eat=1SG.PIV rice AV-come Pawan
 ‘Yesterday I was about to eat when Pawan came.’ prospective [Seediq]
- b. *d(em)a-deru* [i Atrung] *dra patraka*
 ⟨AV⟩RED-cook PIV Atrung ACC meat
 ‘Atrung is cooking meat.’ progressive [Puyuma; Chen 2023]
- there is a complication where AV forms do not surface with clear CV reduplication in Seediq (34)
 - in the prospective aspect, a verb in AV form occurs with a prefix *m-* instead of infix ⟨*m*⟩
 - PV and LV forms show reduplication, e.g. *p-puq-un* ‘RED-eat-PV’ or *r-rnga-un* ‘RED-talk-PV’
 - an assumption here is that reduplication does occur and AV infixes into the reduplicated material, but that material is then deleted and AV surfaces as a prefix⁵ (Holmer 1996) (35)
- (34) “voice” > Asp_{RED} *Asp_{RED} > “voice”
- a. *?ekan* ‘eat’: ?⟨*m*⟩-?ekan [me-?ekan] *?-?⟨*m*⟩ekan [ʔu-mekan]
- b. *rengo* ‘talk’: *r⟨m⟩-rengo* [mu-rengo] **r-r⟨m⟩engo* [ru-mengo]
- c. *gbeling* ‘dig’: *g⟨m⟩-gbeling* [mugu-beling] **g-g⟨m⟩beling* [gumu-beling]
- (35) a. *?ekan* ‘eat’ → *C-?ekan* ‘RED-eat’ → *C⟨m⟩-?ekan* ‘⟨AV⟩RED-eat’ → *me-ekan*
- b. *rengo* ‘talk’ → *C-rengo* ‘RED-talk’ → *C⟨m⟩-rengo* ‘⟨AV⟩RED-talk’ → *m-rengo*
- unexpectedly, the AV marker itself is reduplicated with vowel-initial roots (36)
 - this on the other hand suggests that “voice” is ordered below Asp_{RED}
 - however, it is possible to maintain the order “voice” > Asp_{RED} if reduplication is not exponed until AV is infixed, perhaps as a repair due to lack of available consonants (Holmer 1996) (37)
- (36) * “voice” > Asp_{RED} Asp_{RED} > “voice”
- a. *osa* ‘go’: **m-o-osa* [mo-osa] *m-m-osa* [mu-mosa]
- b. *aha* ‘go’: **m-a-aha* [ma-aha] *m-m-aha* [mu-maha]
- c. *eyah* ‘come’: **m-e-eyah* [me-eyah] *m-m-eyah* [mu-meyah]
- (37) a. *osa* ‘go’ → *C-osa* ‘RED-go’ → *C⟨m⟩-osa* ‘⟨AV⟩RED-go’ → *m-m-osa*
- b. *eyah* ‘come’ → *C-eyah* ‘RED-come’ → *C⟨m⟩-eyah* ‘⟨AV⟩RED-come’ → *m-m-eyah*
- there is also evidence that “voice” and Voice are ordered above Asp_{RED} in reciprocals (38)
 - when there is intervening material between AV and RED, there is evidence of reduplication
- (38) a. *gaalu* ‘care’ → *C-gaalu* ‘RED-care’ → *p-C-gaalu* ‘REC-RED-care’ → *p-g-gaalu*
- b. *gaalu* ‘care’ → *C-gaalu* ‘RED-care’ → *p⟨m⟩-C-gaalu* ‘⟨AV⟩REC-RED-care’ → *m-g-gaalu*

⁵another analysis of these facts is CV reduplication creates an empty C slot which can be filled by the /m/ of the AV marker

5.4 Appl_{CAUS}

- the **causative** morpheme PAn **pa-* is well attested in Austronesian languages (Blust 2009)
 - in causatives, the pivot that 1) AV selects is the causer and 2) PV selects is the causee
 - assuming that CAUS is an Appl head that introduces the causee argument Pylkknen (2000)
- the AV marker does not typically surface with causative morphology (39)
 - /p/ is expected to delete before /m/ given the consonant deletion rule (§1.2), but this is blocked potentially due to competition and/or avoidance of deleting a contentful morpheme⁶

- (39) a. *talang* ‘run’ → **p**-*talang* ‘run.v_{CAUS}’ → **p**⟨**m**⟩-*talang* ‘run.AV.v_{CAUS}’ → *p-talang* (not **m-talang*)
 b. *osa* ‘go’ → **p**-*osa* ‘go.v_{CAUS}’ → **p**⟨**m**⟩-*osa* ‘go.AV.v_{CAUS}’ → *p-osa* (not **m-osa*)

- Appl_{CAUS}** is ordered **below** “voice” and Asp_{RED} across Formosan languages
 - see Chen 2022 as well for Appl_{CAUS} being hosted below Voice_[-D] in Puyuma
 - 1) AV ⟨**m**⟩ does not infix into the root (40) 2) CAUS *p-* is reduplicated and not the root (41)

- (40) “voice” > Appl_{CAUS} *Appl_{CAUS} > “voice”
- | | | |
|----------------------------|---|--|
| a. <i>talang</i> ‘run’: | p ⟨ m ⟩- <i>talang</i> [pu-talang] | * p - t ⟨ m ⟩- <i>alang</i> [putu-malang] |
| b. <i>osa</i> ‘go’: | p ⟨ m ⟩- <i>osa</i> [posa] | p - m - <i>osa</i> [posa] |
| c. <i>chungi</i> ‘forget’: | p ⟨ m ⟩- <i>chungi</i> [pucu-hungi] | * p - c ⟨ m ⟩- <i>hungi</i> [pucum-hungi] |

- (41) Asp_{RED} > Appl_{CAUS} *Appl_{CAUS} > Asp_{RED}
- | | | |
|-------------------------|--|---|
| a. <i>heya</i> ‘stand’: | p ⟨ m ⟩- <i>p-heyu</i> [pupu-heyu] | * p ⟨ m ⟩- h - <i>heyu</i> [puhu-heyu] |
| b. <i>osa</i> ‘go’: | p ⟨ m ⟩- <i>p-osa</i> [pu-posa] | * p ⟨ m ⟩- o - <i>osa</i> [po-osa] |
| c. <i>talang</i> ‘run’: | p ⟨ m ⟩- <i>p-talang</i> [pupu-talang] | * p ⟨ m ⟩- t - <i>talang</i> [putu-talang] |

6 summary

- (42) vocabulary items: head which hosts it:
- | | | | | |
|---------|---|--------------------|----------------------------|-----------------------|
| a. AV | ⇔ | ⟨ m ⟩ | with infix location of __V | “voice” |
| b. DETR | ⇔ | V- | | Voice _[-D] |
| c. REC | ⇔ | p V- | with reduplication | Voice _[+D] |
| d. STAT | ⇔ | k V- | | v |
| e. PFV | ⇔ | ⟨ n ⟩ | with infix location of __V | Asp _{PFV} |
| f. PSV | ⇔ | RED _{CV-} | | Asp _{RED} |
| g. CAUS | ⇔ | p V- | | Appl _{CAUS} |

- the alternation between actor voice (AV) prefix *m-* and infix ⟨**m**⟩ is phonologically predictable
- the prefix *m-* in anticausative, statives, and reciprocals (Chen 2023) contains a regular AV marker
- the order of affixes in Seediq is: Asp_{PFV} > “voice” > Voice_[±D] > Asp_{RED} > Appl_{CAUS} > **v**

⁶an alternate analysis is that AV is impoverished in the presence of CAUS: AV → ∅ / __ CAUS, though this might not be desirable due to cases where AV and CAUS do co-occur when material intervenes between them, e.g. *m-p-p-rengo* ‘⟨AV⟩REC-CAUS-talk’

open questions:

- what projection hosts “voice” if it sits between Asp_{PFV} and $\text{Voice}_{[\pm\text{D}]}$?
- does the affix ordering apply to right-aligned “voice” affixes, such as PV and LV?
 - this is unfortunately hard to test, but some Formosan languages do have prefixal PV
- what about the AV.FUT form *m-p-* in Seediq? what is the identity of *m-* and *p-*?
- what is the interaction between “voice” and mood (i.e. imperative and subjunctive forms)?
 - the AV marker is consistently absent in AV.IMP forms across Formosan languages

References

- Blust, R. (2009). The Austronesian languages. Pacific Linguistics, Canberra.
- Bobaljik, J. D. (2012). Universals in comparative morphology: suppletion, superlatives, and the structure of words. MIT Press, Cambridge, MA.
- Chen, V. (2020). The derived intransitive in Formosan and its implications for the nature of proto-Austronesian actor voice. Oceanic Linguistics, 59(1-2):59–90.
- Chen, V. (2022). When Philippine-type voice meets Indo-European-type voice: insights from Puyuma. Journal of East Asian Linguistics, 31(4):517–550.
- Chen, V. (2023). When sound change obscures morphosyntax: insights from Seediq. Oceanic Linguistics, 62(2):289–303.
- Holmer, A. J. (1996). A parametric grammar of Seediq, volume 30. Lund University Press.
- Kalin, L. (2022). Infixes really are (underlyingly) prefixes/suffixes: evidence from allomorphy on the fine timing of infixation. Language, 98(4):641–682.
- Kastner, I. (2017). Reflexive verbs in Hebrew: deep unaccusativity meets lexical semantics. Glossa, 2(1).
- Kaufman, D. (2012). Predicate classes and PAN* ka. In AFLA-19, pages 26–30, Academia Sinica.
- Labelle, M. (2008). The French reflexive and reciprocal se. Natural Language & Linguistic Theory, 26(4):833–876.
- Pylkknen, L. (2000). What applicative heads apply to. In Proceedings of the 24th Annual Penn Linguistics Colloquium, page 16.
- Ross, M. (1995). Reconstructing Proto-Austronesian verbal morphology: evidence from Taiwan. In Li, P. J.-K., Tsang, C.-W., Ho, D.-A., Huang, Y.-K., and Tseng, C.-Y., editors, Austronesian Studies Relating to Taiwan, pages 727–791. Academia Sinica.
- Ross, M. (2015). Reconstructing Proto-Austronesian verb classes. Language and Linguistics, 16(3):279–345.
- Schafer, F. (2008). The syntax of (anti-) causatives. John Benjamins, Amsterdam.
- Teng, S. F.-c. (2014). Grammaticalization of predicative possession in Nanwang Puyuma and as a basis for reconstruction in PAN. Oceanic Linguistics, 53(1):136–154.
- Travis, L. d. (2010). Inner aspect, volume 80 of Studies in Natural Language and Linguistic Theory. Springer Netherlands, Dordrecht.
- Tsuchida, S. (1976). Reconstruction of Proto-Tsuoic phonology. Institute for the Study of Languages and Cultures of Asia and Africa, Tokyo University of Foreign Studies, Tokyo.
- Tsukida, N. (2005). Seediq. In Adelaar, K. A. and Himmelmann, N., editors, The Austronesian languages of Asia and Madagascar. Routledge, London.
- Yang, H.-f. (1976). The phonological structure of the Paran dialect of Seediq. Bulletin of the Institute of History and Philology, Academia Sinica, 47(4):611–706.
- Yu, A. C. L. (2003). The morphology and phonology of infixation. PhD dissertation, University of California, Berkeley.